

Extraction-Free Scalable Sample Preparation for the Rapid Detection of HIV in Blood Plasma

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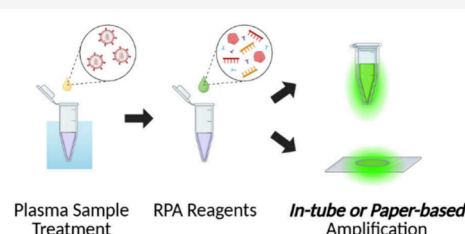


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ABSTRACT: Nucleic acid amplification tests that are rapid and low-cost are needed for expanding access to point-of-care (POC) diagnostics for infectious diseases and viral load monitoring. Several companies have commercialized POC devices using rapid isothermal amplification; however, these tests use nasal swabs or saliva samples. Translating these tests to blood plasma is challenging due to clotting factors, heat- or pH-induced coagulation of human albumin, high concentrations of ribonucleases (RNases), and other sample matrix inhibitors. Crude, extraction-free sample preparation methods are promising alternatives that reduce the complexity of sample preparation. In this paper, we present an extraction-free sample preparation method for in-tube and paper-based reverse transcription recombinase polymerase amplification for the detection of HIV in blood plasma samples. We use a rapid heat treatment protocol with a reducing agent to inhibit RNases and lyse HIV virus, enabling RNA preservation into direct amplification without the need for complex RNA purification steps. We demonstrate sample treatment in 5 min and amplification within 20 min in reaction volumes ranging from 50 to 300 μ L and show that large volume reactions achieve a limit of detection of 800 copies/mL (16 copies/reaction) for HIV-positive plasma in both paper and tube formats.



INTRODUCTION

Nucleic acid amplification tests (NAATs) are crucial tools for diagnostics and disease monitoring due to their high clinical sensitivity and specificity. NAATs are typically conducted in centralized laboratories using expensive automated platforms capable of high-throughput testing. This centralized approach is driven by the requirement of thermocycling, high cost instrumentation, cold chain logistics for reagents, trained staff, and infrastructure to store required equipment.¹ Point-of-care (POC) diagnostics aim to provide low-cost, rapid results at the location of patient care to alleviate these obstacles for widespread access and dissemination.^{1,2} To guide the development of POC technologies, the WHO established the ASSURED criteria (Affordable, Specific, Sensitive, User-friendly, Rapid and robust, Equipment-free, and Delivered) to describe the ideal point-of-care diagnostic.³ More recently, the REASSURED criteria has been proposed, adding “Real-time connectivity” and “Ease of specimen collection” to the original framework.⁴

In response to the need to meet the REASSURED criteria, several POC NAAT platforms have been commercialized, including Abbott *m-PIMA*, Diagnostics for the Real World *SAMBA II*, and *Enigma ML*. These platforms are designed to simplify user steps and minimize the need for specialized training, with many receiving CLIA-waived status.⁵ Despite their accuracy and relative ease of use for decentralized testing, their long runtimes and high upfront costs, ranging from \$10,000–25,000, remain a significant barrier to their adoption at primary and community healthcare levels.^{6,7} More recently,

home-based NAAT testing has expanded with companies like Cue Health, Lucira, and Detect commercializing POC diagnostic tests using isothermal NAATs for SARS-CoV-2 and Flu. While these home-use devices are less expensive than the POC platforms, they are still costly at \$65–90 per test cartridge.^{8–10} One of the key factors enabling their viability is that they use saliva or nasal swab samples for respiratory infections, which are easier to collect, use simplified sample preparation, and contain high target concentrations.

Blood is an essential sample type for molecular testing, as many pathogens can be found suspended in plasma or residing in host cells, and is a relatively consistent sample.¹¹ Blood samples pose major challenges to POC NAATs because they require stringent and complex sample preparation, can contain lower target concentrations, and include numerous inhibitors to nucleic acid amplification assays and nucleases that can degrade target strands.¹² As a result, sample preparation has long been a bottleneck in POC test development.¹²

Sample preparation typically requires pathogen lysis, denaturation of DNases and RNases, and the purification of target DNA or RNA from sample inhibitors.^{13,14} Despite

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providing highly purified nucleic acid targets for amplification, solid phase extraction and similar protocols can be difficult to implement in a POC environment due to the need for multiple buffer exchanges, requirement of concentrated toxic chaotropic salts that can inhibit amplification, and long runtimes of 30–60 min.¹³ In contrast, POC respiratory tests such as Cue and Lucira benefit from simplified sample preparation for saliva and nasal swab samples that avoid many blood-associated challenges, enabling their broader availability. Blood and plasma samples are particularly challenging to use as they contain clotting factors and human albumin that can easily coagulate due to changes in temperature or pH.¹⁵ Additionally, blood-based samples have a high concentration of ribonucleases that degrade exogenous RNA and contain significant NAAT inhibitors like immunoglobulins, lactoferrins, and hemoglobin.^{16,17} Blood-based testing remains a significant challenge for POC NAAT development.^{18,19} Some POC approaches have addressed sample preparation using microfluidic channels, functionalized magnetic beads to manipulate the sample, or slide mechanisms to introduce buffers to a sample membrane.^{20–22} While these devices can reduce some pipetting and centrifuging steps, they still require multiple user interventions to purify and elute the nucleic acids and remove inhibitory agents and often involve long runtimes. Approaches to automating these steps typically incur high costs due to pumps, actuators, and other robotic components required for fluid manipulation.^{20–29} Others have used isotachopheresis (ITP), which implements an electric field and a buffer system, to concentrate and extract nucleic acids.^{30,31} These purification techniques require additional time, user steps, and complexity to the workflow, complicating their use in low-cost sample-to-answer POC diagnostics. Significant efforts in reducing device complexity and the associated high costs for POC NAATs are still needed for blood-based targets.

After sample preparation, the sample undergoes nucleic acid amplification. The gold standard NAAT, polymerase chain reaction (PCR), uses precise thermocycling to facilitate amplification and detection of nucleic acid targets.^{32,33} The complexity of PCR and the instrumentation costs associated with the rapid temperature cycling process typically requires 45–90 min and is challenging to implement in low-cost, point-of-care (POC) NAATs.³² Some laboratories have adopted miniaturized PCR that uses small reaction volumes and fast thermocycling to reduce overall time. These methods remain costly and complex, often relying on intricate chip designs, multiple valving and mixing mechanisms, and expensive equipment.^{23–25} Isothermal NAATs are well-suited for POC settings, as they have similar limits of detection as PCR while operating at a single temperature, thus reducing the requirement of hardware with precise temperature control. Isothermal methods can be performed in a single temperature range of 39–65 °C. LAMP amplification takes 30–60 min at 65 °C to complete and often struggles with poor specificity.³⁴ Recombinase polymerase amplification (RPA) is an isothermal NAAT that detects target nucleic acids in less than 15 min, operates at a temperature range of 37–42 °C, and features high specificity due to its sequence-specific, real-time probe.³⁵ More recently, RPA and other isothermal NAATs have been applied to microfluidic paper-based analytical devices (μ PADs) to reduce complexity and cost, benefiting from paper's ease of stacking, storing, transporting, and ability to dry chemical or biological reagents.^{36–39} Nonetheless, μ PADs often suffer from uneven analyte concentration profiles after rehydration, poor

detection output, and reduced sensitivity when compared to traditional tube-based assays.^{40–44} Strategies such as careful membrane selection, stacking membranes to improve analyte distribution, and implementing mixing mechanisms to redistribute reagents have improved assay performance.^{40,45,46} Despite these advances, further development is needed to create rapid and reliable POC NAAT devices that integrate both sample preparation and amplification.

Crude, extraction-free sample preparation is an emerging strategy that reduces the workflow complexity and cost for POC NAATs. By using lysis and nuclease inactivation reagents that are compatible with NA amplification, users can amplify targets without the need for extensive sample preparation.⁴⁷ Extraction-free protocols for DNA and RNA targets have been demonstrated on a variety of samples such as nasal swabs, saliva, urine, and blood.^{47–52} Curtis et al. developed an extraction-free protocol that adds minimally processed diluted whole blood directly into RT-LAMP.⁴⁹ More recently, Wang et al. developed an extraction-free method for serum and plasma samples by directly adding the sample into RT-LAMP and improving assay sensitivity by increasing reaction volume.⁵³ The direct sample addition approach by Wang et al. presents challenges for lower-temperature isothermal NAATs like RPA, as it does not provide sufficient heat required for complete pathogen lysis or ribonuclease (RNase) inhibition, which is problematic at low target concentrations. Additionally, the crude sample preparation approach is more effective for respiratory and saliva sample types, which contain fewer and less potent nucleic acid amplification inhibitors and coagulation factors but remain challenging for viruses, like HIV, that have low limit of detection (<1000 copies/mL) requirements.^{54,55} While some studies have demonstrated successful implementation of extraction-free sample preparation in blood products, their performance remains limited.^{50,56} For example, Myhrvold et al. achieved detection of Zika RNA in blood and serum at a limit of detection of $\sim$ 45,000 copies/mL.⁵⁰ These reports highlight that extraction-free approaches are feasible, but further improvements are required to improve the limit of detection (LoD), streamline user steps, and enhance the robustness of target RNA preparation from blood-based samples.

In this paper, we present an extraction-free sample preparation method for in-tube and paper-based reverse transcription recombinase polymerase amplification (RT-RPA) for the detection of HIV in blood plasma samples. We use a rapid heat and dithiothreitol treatment protocol that inhibits RNases and lyses HIV virus to enable effective RNA preservation and direct amplification without the need for complex purification steps. We then directly added the incubated sample into RT-RPA for paper-based or in-tube reactions. We demonstrate sample treatment in 5 min and amplification within 20 min. We perform RPA reactions in reaction volumes ranging from 50 to 300 μ L and show that large volume reactions achieve a limit of detection (LoD) of 800 copies/mL (16 copies/reaction) for HIV-positive plasma in both paper and tube formats, highlighting the potential for scalable, user-friendly, and sensitive molecular diagnostics for blood-borne diseases in resource-limited settings.

MATERIALS AND METHODS

Biological Samples

We used pooled human plasma (BioIVT, Westbury, NY, USA) and a cultured HIV-1 subtype B (8E5) linearity panel stored in plasma (SeraCare, Milford, MA, USA). The 8E5 HIV virus contains a single base pair addition in the pol gene of the genome, resulting in a reverse transcription-defective virus.

Sample Preparation Protocol and RNase Activity Measurements

To enable extraction-free detection of HIV-positive plasma samples, we used dithiothreitol (DTT) and thermal treatment to inactivate ribonucleases. The structure of RNases is stabilized by disulfide bonds.⁵⁷ Reducing agents such as DTT break disulfide bonds, effectively destabilizing RNases and inhibiting their activity.⁵⁸ The addition of heat further relaxes the RNase structure, allowing DTT to more effectively break down disulfide bonds while concurrently lysing HIV virions.⁴⁷ Pooled human plasma is diluted 1:3–1:3.33 in nuclease-free water containing 7.5–7.7 mM dithiothreitol (DTT, Sigma-Aldrich, MA, USA). The diluted sample is incubated at 70 °C for 5 min, achieving viral lysis and inactivation of plasma-derived RNases.

To evaluate the sample preparation protocol, we used an RNaseAlert Substrate Detection system (Integrated DNA Technology, Coralville, IA, USA) and varied the DTT concentration as well as incubation temperature. The assay employs a short RNA oligonucleotide with fluorescein and a quencher on either end. Upon cleavage, the fluorophore is separated from the quencher, resulting in a measurable green fluorescence signal. The experiments were conducted in a black 96-well plate (M9685, Corning Inc., Corning, NY, USA) and sealed with PCR film before testing. Each well contained 1× RNase Alert buffer, 10 μ L of RNaseAlert substrate, 26 μ L of sample, and water to reach a final volume of 100 μ L. Negative sample controls consisted of water only, while positive controls contain 5 μ g of RNase A (ThermoScientific, MA, USA). The plates are heated to 37 °C, and fluorescence was monitored to determine residual RNase activity under various treatment conditions using a plate reader (SpectraMax iD3, Molecular Devices, San Jose, CA). An initial 10 s orbital mix step is performed before running the test for 30 min. The excitation and emission wavelengths are set at 485 and 535 nm, respectively, with the photomultiplier tube gain set at a 140 ms exposure.

Tube-Based RT-RPA Reactions

For sample preparation of 50 μ L reactions, 3 μ L of HIV-positive plasma is added to 7.7 mM DTT and nuclease-free water at a 1:3.33 ratio of plasma to diluent. The sample is incubated at 70 °C for 5 min on a heat block (10153–318, VWR, US). All 13 μ L of the diluted sample is then added to the RT-RPA mastermix to reach a final 50 μ L volume, as shown in Figure 1. We incubated the RT-RPA reactions at 39 °C for 20 min in an isothermal fluorescence reader (T16-ISO VIC, Axxin, AU) to capture real-time RT-RPA fluorescence reactions. We perform a vigorous manual shaking step after 5 min of incubation before placing tubes back in the reader for the rest of the 15 min, as per TwistDx instructions.⁶² In 50 μ L reactions, the RT-RPA mastermix consists of a TwistAmp exo kit lyophilized pellet (TwistDx, UK), 29.5 μ L of rehydration buffer, 14 mM magnesium acetate, 540 nM forward and reverse primer (Integrated DNA Technologies, USA), 120 nM exoprobe (LGC Biosearch Technologies, GBR), and 0.5 U μ L⁻¹ of reverse transcriptase (Agilent AffinityScript, USA). We use HIV-1 primers and probe for RT-RPA reactions developed by Lillis et al.⁵⁹ We included 1% w/v Triton-X and 0.04% w/v BSA additives to the RT-RPA mastermix to help stabilize enzymes from amplification inhibitors present in plasma samples.^{60,61}

We also perform experiments in larger, 300 μ L, reactions to achieve lower limits of detection.^{53,63} For sample preparation of 300 μ L reactions, 20 μ L of HIV-positive plasma is added to 7.5 mM DTT and nuclease-free water at a 1:3 ratio of plasma to diluent. The sample is incubated at 70 °C for 5 min and added to the RT-RPA mastermix.

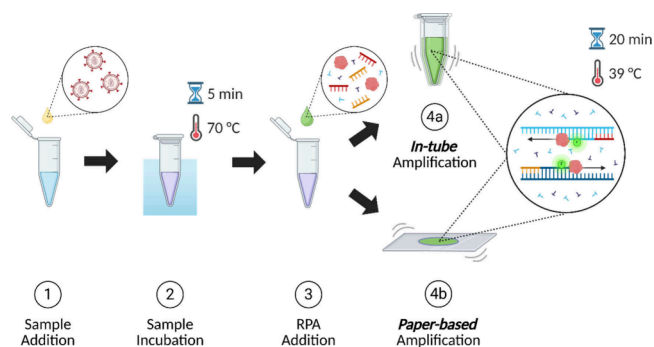


Figure 1. Overview of the extraction-free sample preparation for HIV in plasma. First, plasma is added to the diluted DTT solution. The sample is incubated for 5 min at 70 °C and added to the RT-RPA reaction solution. The reaction is incubated at 39 °C for 20 min on a benchtop fluorometer or in a paper membrane set on a custom vibration mixing and heating platform (see Figure S1) and imaged with a fluorescence microscope. Tube-based reactions include a vigorous mix step after 5 min of incubation as recommended.⁶² For paper-based reactions, continuous vibration-based mixing is done on a mixing and heating platform to continuously redistribute RPA reagents and improve paper-based reaction efficiency.⁴⁰

All 80 μ L of the incubated sample is added to the mastermix to reach a 300 μ L reaction volume. This plasma concentration is slightly higher than that used in 50 μ L reactions to simplify dilution calculations across steps. Experimentally, 6–8% plasma shows comparable performance in RPA reactions (Figure S3), and this modest increase from 6 to 6.7% did not affect the extraction-free results. In 300 μ L reactions, the RT-RPA mastermix consists of 6 TwistAmp exo pellets, 177 μ L of rehydration buffer, 14 mM magnesium acetate, 540 nM forward and reverse primer, 120 nM exoprobe, 1% w/v Triton-X, 0.04% w/v BSA, and 0.5 U μ L⁻¹ of reverse transcriptase. These in-tube reactions are performed at 39 °C in the plate reader. The 300 μ L RT-RPA experiment is evenly split into two wells of a black 96-well plate. A 2 mm steel ball (TwistDx, UK) is added to each reaction well and sealed with PCR film. A 40-s orbital mix step at the highest setting is done before running the test and after 5 min of incubation. To generate results, we take the average of the two wells and plot the bulk fluorescence against time.

Paper-Based RT-RPA Reactions

In 50 μ L paper-based RT-RPA experiments, we add incubated HIV sample, magnesium, and RT-RPA mastermix (prepared as described above) onto a circular 9.21 mm diameter GF/DVA pad (1835–6621, Whatman, UK) that is supported by a 2 in. $\times$ 3 in. glass slide (CA6101, Premiere, CN). We manually shake the reaction in the tube for 20–30 s to ensure homogeneous distribution of reactants before evenly pipetting the reaction onto the circular pad and sealing it with PCR film (TempPlate RT Select Optical Film, USA Scientific, USA). The glass slide is slotted into the mixing device with the PCR film in contact with the motor. The vibration motor uses a 2 V input, and the PTC uses a 2–3 V input.

For 300 μ L paper-based reactions, we add incubated HIV sample, magnesium, and mastermix onto a 20.55 mm diameter GF/DVA circular pad that is again supported by a 2 in. $\times$ 3 in. glass slide. We manually shake the reaction in the tube before evenly pipetting it onto the circular pad and sealing it with the PCR film. The glass slide is slotted into the mixing platform with the PCR film in contact with the titanium foil circle. The motor uses a 2.4 V input, and the PTC is set at a 3–4 V input.

For paper-based reactions, we use a vibrational mixing device that we reported on in previous work, to provide continuous mixing of the RT-RPA reaction and enhance the amplification.⁴⁰ Using the vibration platform, we see a significant improvement in paper-based reactions for this extraction-free method, Figure S2. In brief, the mixing device consists of a 3D printed assembly with an eccentric

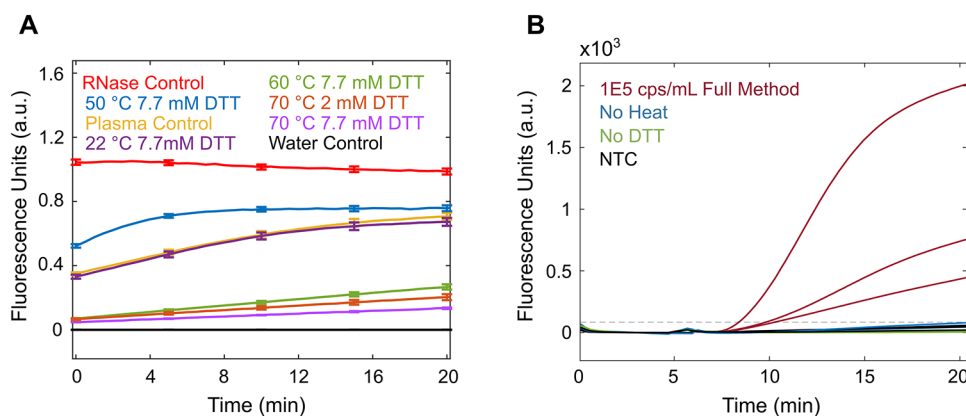


Figure 2. We monitored RNase activity of plasma samples after a 5 min, heated incubation with DTT to determine the most effective DTT concentration and incubation temperature required to deactivate RNases in plasma. (A) Fluorescence of the RNase Alert assay is plotted as a function of time for various DTT concentrations and incubation temperatures. Sufficient RNase inhibition was observed with 7.7 mM DTT incubated at 70 °C for 5 min. Without the incubation method, no thermal treatment and no DTT, there is little to no RNase inhibition. For all experiments, $n = 3$ with error bars representing standard error. (B) We performed RT-RPA reactions of samples that underwent extraction-free sample preparation with both thermal and DTT treatment, without DTT treatment, or without thermal treatment. Using the full method, we show positive amplification for 1E5 copies/mL HIV plasma samples. Without the full initial incubation method (i.e., no heat, or no DTT), little to no amplification is observed due to the degradation of the target HIV-1 RNA by RNases. The dashed line represents the threshold.

rotating mass vibrational motor (VC1234B016F, Vybronic, CN), PTC heater (PTC Heating Element, DFRobot, CN), and a rubber isolation foam (Thermwell, USA). The vibration assembly positions the RPA reaction pad, which is supported by a glass slide and PCR film, in contact with the motor and PTC to provide sufficient mixing and heating for RPA reactions, as shown in Figure S1. This device is used for both 50 and 300 μ L paper-based reactions. For 50 μ L reactions, the RPA pad and PCR film are in direct contact with the vibrational motor. For 300 μ L reactions, we cut a 0.12 mm thick titanium foil (LiteOutdoors, CAN) into a 25 mm diameter circle and adhered it to the top of the vibration motor using epoxy adhesive (Loctite, USA). The titanium foil supports the larger 300 μ L reactions, providing even heating during RT-RPA and distributing vibrational mixing to the entire circular pad. Without the foil, mixing and heating would be localized to the 12 mm diameter of the motor.

Imaging and Data Analysis

We imaged paper-based reactions using an epifluorescence microscope (AZ100, Nikon, JPN) and illumination system (X-Cite exacte, Excelitas Technologies, USA) with a 1X objective and 35 mm field of view. We use an epifluorescence filter cube set (XF100-2, Omega Optical, LLC., USA) and CMOS camera (Prime BSI Express, Teledyne, Photometric, USA) to capture grayscale images every 5 s for 20 min. We use an image analysis algorithm (MATLAB, MathWorks, USA) to calculate the fluorescence intensity over the full 20 min. For the analysis of membrane reactions, we imaged the entire pad area. We set a region of interest as the active membrane. For 50 μ L reactions, the ROI was a 9.2 mm diameter circle, and for 300 μ L reactions, it was 20.6 mm. We perform a background subtraction to eliminate autofluorescence from the reaction pad, PCR film, glass slide, and initial nonspecific assay interactions. For each specific time point, we calculated the total fluorescence by summing the intensities over the ROI. The intensity over the entire circular pad is averaged and plotted against time. To determine the threshold between positive and negative reactions for all experiments, we set it as the average intensity of the no template control (NTC) experiments plus three standard deviations. We used this threshold for all amplification figures.

RESULTS AND DISCUSSION

We demonstrate a simple, rapid, and extraction-free sample preparation strategy for tube and paper-based RT-RPA. For this extraction-free approach, diluted pooled blood plasma is incubated with DTT. The thermal and chemical process

inhibits ribonucleases (RNases) as well as lyses HIV virions. The resulting plasma lysate is then directly added to RT-RPA for tube- or paper-based amplification. We validate the sample preparation method for the inactivation of RNases using the RNase Alert assay. We show that HIV can be amplified in tube- or paper-based formats, and we demonstrate that this approach exceeds the recommended LOD requirements for HIV-1 testing using rapid, extraction-free sample preparation.

Sample Preparation Validation

We first focus on demonstrating that we can lyse HIV virions and inactivate RNases in a short, heated incubation step. To achieve this, we studied the RNase inactivation step, as there are high concentrations of RNase enzymes in blood plasma that degrade exogenous RNA.¹⁶ We studied various DTT concentrations and heat incubation temperatures to develop a sample preparation method that sufficiently inhibits the RNase activity in plasma samples. We then paired this sample preparation method with 50 μ L of tube-based RT-RPA reactions for detecting HIV from plasma.

Figure 2A shows the RNase Alert activity of plasma lysates under various DTT concentrations and incubation temperatures. When RNases are present, the template RNA oligonucleotide is cleaved and the fluorescence intensity increases rapidly with time. The RNase control experiments represent the maximum fluorescence in the assay or what would result for experiments with strongly active RNases. Note that a significant increase in fluorescence occurs prior to time zero because there is finite time required to mix the components and begin the fluorescence read. We examined incubation temperatures ranging from 22 to 70 °C and DTT concentrations between 2 and 7.7 mM (final concentration of 0.5–2 mM DTT). The results show that incubation at higher temperatures improves RNase deactivation, resulting in lower fluorescence slopes. At higher incubation temperatures of 70 °C, increasing DTT concentration from 2 mM to 7.7 mM only marginally improves the deactivation of the RNases (see DTT concentrations up to 23.1 mM in Figure S5). At temperatures greater than 70 °C we observed sample coagulation that inhibited pipetting and sample manipulation. Additionally,

significant RNA degradation is known to occur at incubation temperatures above 70 °C.⁶⁴ To mitigate sample coagulation while effectively inhibiting RNases, we set the plasma-to-buffer ratio to roughly 1:3 and identified 70 °C as the minimum temperature required for a 5 min incubation period. For the buffer, we used nuclease-free water with 7.7 mM DTT to inhibit the RNases. In these experiments, the 1:3 dilution ratio resulted in a total reaction concentration of 6% plasma. At this concentration, the assay provided consistent results and represents the maximum sample volume that the assay can accept with the dilution step. We consistently observed high RNase activity for the 5 min incubation setting at 50 °C. To mitigate this while maintaining a relatively low incubation temperature, longer incubation times with higher reducing agent concentrations may be necessary.

We evaluated this optimal extraction-free plasma sample preparation protocol in 50 μ L, tube-based HIV-1 RT-RPA reactions to demonstrate its compatibility with RT-RPA, as shown in Figure 2B. Using a 5 min incubation at 70 °C with 7.7 mM DTT, we observe positive amplification for 1E5 copies/mL of HIV-1 plasma. Without DTT or 70 °C incubation, we observe little to no fluorescence for 1E5 copies/mL HIV plasma. These experiments demonstrate the compatibility of the sample preparation method with RT-RPA for HIV-positive plasma as well as the need for both DTT and elevated incubation temperatures to adequately lyse virions and inactivate RNases. Some residual RNase activity remains after incubation; however, with subsequent dilution of the sample into the RT-RPA reaction, the low RNase activity does not prevent amplification. This residual activity along with plasma inhibitors and varying degrees of RNA degradation during the incubation period may explain variations in the amplification curves for a given concentration. We do not address plasma separation in this work, but this step is necessary for processing whole blood samples. Plasma extraction is typically achieved through centrifugation; however, for point of care applications, low-cost plasma separation membranes and commercial membrane-based devices have become available.^{65–68} In future work, we aim to integrate this step along with sample metering to develop a more comprehensive, extraction-free plasma sample preparation protocol.

50 μ L Paper-Based RT-RPA Reactions

We evaluated the LoD for standard 50 μ L RT-RPA reaction volumes with the extraction-free sample preparation protocol for HIV-1 in plasma in paper-based RT-RPA reactions performed on our custom vibration mixing and heating platform. We implemented our mixing and heating platform to improve the performance of paper-based reactions and maximize the assay sensitivity. In our previous work, we showed that vibration mixing significantly improved paper-based RPA reactions by continuously redistributing reagents for the reaction, as seen in Figure S1. With this mixing mechanism, we found an overall fluorescence output increase up to 16-fold and up to a 50% reduction in time to threshold when compared to unmixed experiments for HIV DNA.⁴⁰ With the mixing platform, we see results similar to those with the extraction-free method, as seen in Figure S2. Figure 3A shows RT-RPA fluorescence images at $t = 20$ min for no template control, 1E3, 1E4, and 1E5 copies/mL plasma samples, with no background subtraction added. The results show that without HIV-1 target present, there is minimal

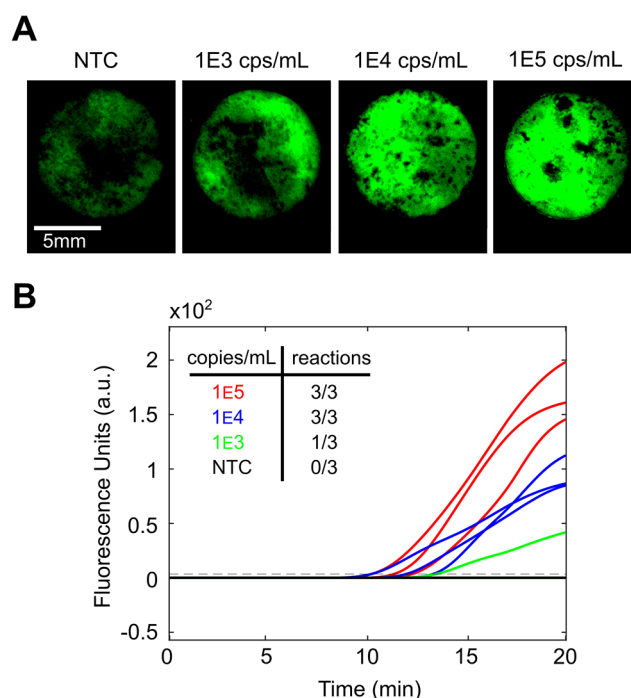


Figure 3. (A) Epifluorescence images of 50 μ L, paper-based, RT-RPA reactions for no template control, 1E3, 1E4, and 1E5 copies/mL of HIV-1 positive plasma using the extraction-free sample preparation protocol at $t = 20$ min. These images have no background subtraction added and use the same brightness and image contrast settings. To see images with background subtraction, see Figure S4. (B) Total fluorescence amplification curves for 50 μ L paper-based reactions for $n = 3$. For these experiments, the LoD is 1E4 copies/mL or 30 copies/reaction with all three reactions showing positive amplification. At 1E3 copies/mL, only one reaction was positive and for all NTCs, none of the reactions amplified. Data includes initial background subtraction to remove background autofluorescence.

fluorescence, mainly autofluorescence due to the reaction pad, PCR film, and initial nonspecific interactions in RT-RPA. The autofluorescence is initially observed at $t = 0$ min for both NTCs and positive samples and remains constant throughout the entire experiment for NTC trials. When HIV-1 is present in the sample, we observe strong fluorescence intensity, indicating positive amplification. With background subtraction, we found our membranes to be completely dark for NTCs and had fluorescence patches in positive experiments, as seen in Figure S4.

Figure 3B depicts the total fluorescence for paper-based RT-RPA reactions as a function of time for HIV-positive plasma samples and NTCs. Fluorescence remains constant at the baseline level until 10 min when it rapidly increases, consistent with exponential amplification. The fluorescence intensity increase appears later and is inhibited relative to typical paper-based RT-RPA experiments conducted in clean (no plasma) samples at similar HIV-1 concentrations. We observe that the fluorescence intensity grows more rapidly with time as the number of target copies increases. The average end point fluorescence at $t = 20$ min also increased with concentration. The threshold is determined as the average intensity of the no template control (NTC) experiments plus three standard deviations. We observed an LoD for 50 μ L of paper-based reactions of 1E4 copies/mL or 30 copies/reaction. The LoD of 1E4 copies/mL does not meet WHO HIV-1 testing

recommendations of 1,000 copies/mL,⁵⁴ and so we explored larger volume RPA experiments with the goal of lowering the LOD to meet WHO recommendations.

Improving the Limit of Detection Using Large-Volume RT-RPA Reactions

To improve the LoD, we adapted the assay to accept larger volumes of plasma sample by increasing the RT-RPA reactions to a 300 μ L reaction volume. Larger volume RPA reactions have been shown to improve assay analytical sensitivity by accommodating larger sample volumes and increasing dilution to mitigate challenges with coagulation and sample inhibitors.⁶³ Previous research by Wang et al. showed that 500 μ L RT-LAMP reactions maintain the analytical sensitivity, in terms of copies per reaction, as the typical 20 μ L RT-LAMP reactions, while accepting a larger volume (20 μ L) of serum, effectively resulting in improvements in the LoD (copies/mL).⁵³ We use this large volume amplification approach to improve the LoD by accepting 20 μ L plasma samples, within the range of a typical fingerstick sample.⁶⁹

We evaluated the extraction-free plasma sample preparation strategy for 300 μ L tube-based RT-RPA reactions. Figure 4

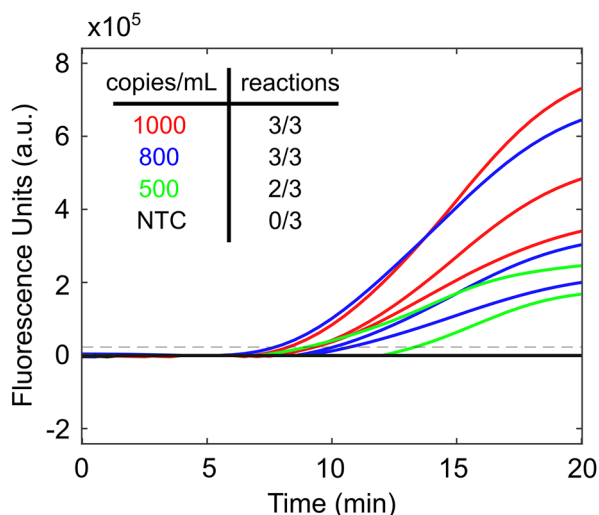


Figure 4. Bulk fluorescence curves from 0 to 1000 copies/mL of HIV-1 positive plasma for 300 μ L in-tube RT-RPA reactions at the extraction-free sample preparation method ($n = 3$). The LoD for these reactions was 800 copies/mL or 16 copies/reaction. At 500 copies/mL, two of three replicates amplified within the 20 min span of the assay.

shows the bulk, tube-based fluorescence curves for 300 μ L of RT-RPA reactions as a function of time for HIV-1 positive plasma samples. With increasing concentrations of HIV-positive plasma the maximum fluorescence at $t = 20$ min increases. At 1E5 cps/mL, the average end point fluorescence is $\sim 4.9\text{E}5$ a.u. and for 1E4 cps/mL, $\sim 3.6\text{E}5$ a.u. For 500 copies/mL, only two of three reactions amplified. As expected, the NTC or 0 copies/mL reactions did not amplify. For these large-volume tube-based reactions, we achieved a limit of detection of 800 copies/mL or 16 copies/reaction, exceeding the WHO LoD recommendation of 1000 copies/mL.

Paper-based NAAT reactions typically perform worse than tube-based reactions resulting in reduced amplicon yield and poor limits of detection.^{41–43,45} Here, we perform 300 μ L of RT-RPA reactions in glass fiber membranes using the vibration platform to provide continuous mixing and a constant 39 $^{\circ}\text{C}$

incubation temperature. Figure 5A shows the fluorescence images for 300 μ L paper-based RT-RPA reactions at $t = 20$

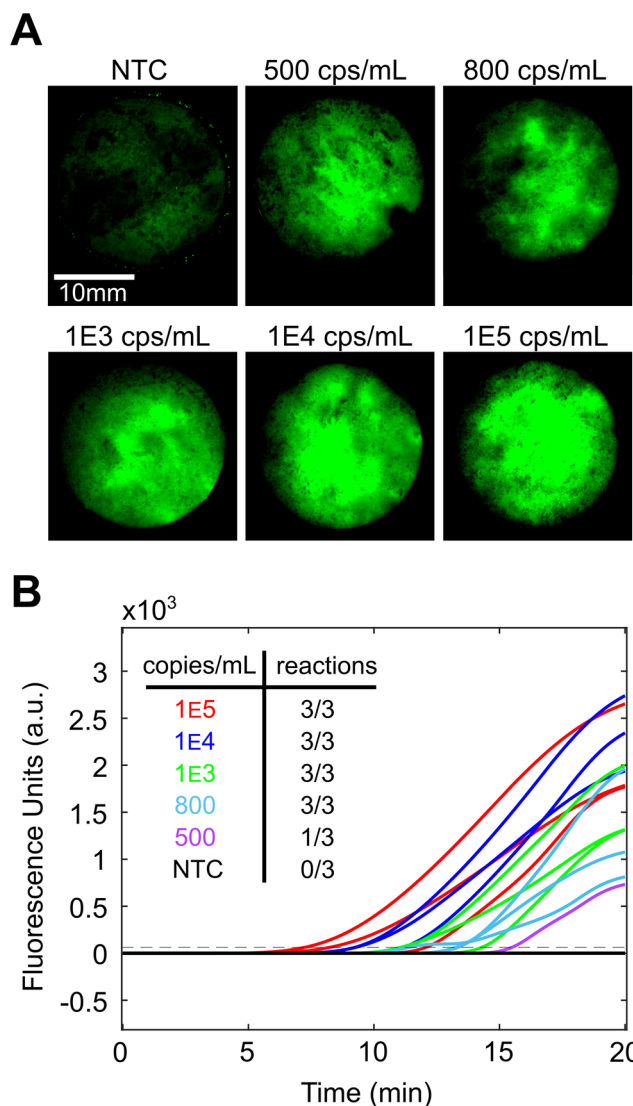


Figure 5. Epifluorescence images and total fluorescence curves for 300 μ L paper-based RT-RPA reactions using extraction-free sample preparation from 0 to 1E5 copies/mL of HIV-1 positive plasma. These images have no background subtraction added and use the same brightness and image contrast settings. To see images with background subtraction, see Figure S5. (A) Epifluorescence images of 300 μ L volume RT-RPA reactions in glass fiber pads at $t = 20$ min for all tested concentrations of 0–1E5 copies/mL. (B) Total fluorescence curves for 300 μ L paper-based reactions, at $n = 3$. We tested concentrations from 0 to 1E5 copies/mL of HIV positive plasma and found the LoD to be 800 copies/mL or 16 copies/reaction. At 500 copies/mL, one of three reactions was positive. The text color for copies/mL correlates to the same color on the plot. Data includes initial background subtraction to remove background autofluorescence.

min for all tested concentrations of 0–1E5 copies/mL. Like the 50 μ L experiments, there is no background subtraction applied, so the NTC shows slight autofluorescence. With increasing concentrations of HIV, we see brighter fluorescence and a larger distribution of fluorescence, indicating positive amplification.

Figure 5B shows the area-averaged fluorescence curves for large-volume, paper-based RT-RPA reactions using HIV-positive plasma. We tested a range from 0 to 1E5 copies/mL of positive HIV plasma. For 800–1E5 copies/mL, all reactions amplified. At 500 copies/mL, one reaction amplified after $t = 14$ min with two showing no amplification. In comparison, tube-based reactions showed slightly better performance with two positive reactions at 500 copies/mL and shorter times to threshold for 800 and 1000 copies/mL. The times to threshold of paper-based reactions were a minute and a half later compared to in-tube experiments. This work demonstrates that extraction-free sample preparation of blood plasma is compatible with both paper and tube-based RT-RPA reaction approaches. Additionally, we further reduce user steps and device complexity that is associated with solid phase extraction methods and can achieve rapid HIV detection with a full workflow under 25 min.

SUMMARY

We present extraction-free sample preparation for in-tube and paper-based RT-RPA detection of HIV-positive plasma. We demonstrate a low-complexity workflow for HIV detection from human plasma that incorporates viral lysis, RNase inactivation, and direct addition of crude lysate into a rapid RT-RPA assay, achieving a total runtime of 25 min (5 min for sample treatment and 15–20 min for amplification). We demonstrate scalable RT-RPA reactions, from 50 to 300 μ L. Larger reaction volumes enable larger plasma sample volumes to be added and result in lower LoDs. As a result, we achieve an LoD of 800 copies/mL (16 copies/reaction) of HIV positive plasma for both tube- and paper-based RT-RPA reactions. Previous examples for the detection of HIV via extraction-free NAAT approaches have used LAMP for amplification, which requires a runtime of 45–60 min.^{49,53,70} To our knowledge, this is the first example of extraction-free HIV detection from plasma using RT-RPA, leveraging its rapid runtime (15 min) and low incubation temperature (40 °C). This method demonstrates that laborious nucleic acid extraction of HIV RNA from blood plasma samples can be avoided by adding minimally processed lysate to RT-RPA reactions for rapid detection. This approach may assist in simplifying NAAT detection platforms and, with further development, could improve accessibility to the rapid and affordable detection of blood-borne viruses.

While 300 μ L large volume reactions achieve a lower LoD, this increases the reagent costs 6-fold compared to typical 50 μ L RT-RPA reactions. Bulk manufacturing and lyophilization will likely reduce costs, but for assays with acceptable LoDs, small reaction volumes are possible. For 50 μ L reactions, we achieve an LoD of 1E4 copies/mL. For targets such as Hepatitis C or Hepatitis B, the acceptable LoD is significantly higher at 3,000 IU/mL and 1E4–1E5 copies/mL, respectively.^{71,72} For these samples, we anticipate that 50 or 100 μ L of reactions would be sufficient to reach these LoDs.

Microfluidic paper-based analytical devices (μ PADs) are popular due to their application in POC NAATs. Paper is low-cost and conducive to manufacturing, storage, transportation, stacking, and deposition and lyophilization of reagents.^{36,37} Paper-based devices are an exciting approach for low-cost, fully integrated, self-contained, and minimally instrumented diagnostics that are required for POC NAAT deployment.^{38,73–76} Previous research has shown worse RPA performance on membranes, which has limited its deployment in paper-based

NAATs.⁴¹ We found that with continuous mixing, paper-based and in-tube RT-RPA reactions had the same LoD for the extraction-free sample addition method in large 300 μ L volume reactions. Although the current method does not encompass a fully paper-based workflow, the reported results do demonstrate that extraction-free sample preparation and RT-RPA on membranes can achieve sensitive detection of HIV from blood plasma. This demonstrates feasibility for adapting this method to a fully paper-based, low-cost POC HIV diagnostic. For future work, sample treatment will be developed to additionally work on paper. The current method excludes the plasma extraction step, as the HIV samples were received and stored in plasma. Future work will investigate plasma separation from whole blood using low-cost membranes or other alternative methods. While large-volume reactions meet the sensitivity threshold recommended by the WHO, they do not yet reach the more stringent CDC guideline of 200 copies/mL.⁷⁷ Additional efforts to improve the assay may focus on improving sensitivity, reducing reaction volumes by increasing plasma concentration added, and extending the extraction-free approach to other molecular targets such as Hepatitis B and C.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.5c05443>.

Schematic and images of vibration platform, membrane-based RT-RPA results under mixed and unmixed conditions, plasma inhibition studies using synthetic DNA, and additional time series images of paper-based RT-RPA reactions (PDF)

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review and editing, supervision. Coleman Martin: conceptualization, methodology. Nandini Gummalla: investigation, methodology. Jonathan D. Posner: conceptualization, writing - review and editing, supervision, project administration, funding acquisition.

Notes

The authors declare no competing financial interest.

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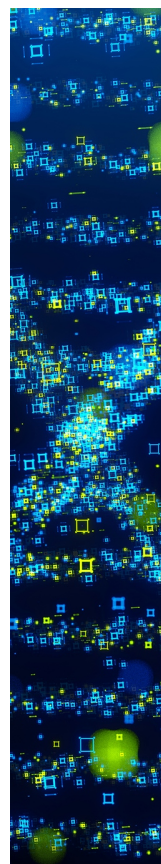
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